Jacobs Building Redesign

MAE Enterprise

Head of Project Management Alex Duan
Head of Finance Malek Gallichio
Head of Design Elizabeth Vazquez

EG 1003 Section A2 | Final Presentation | April 29, 2021

Agenda

- Problem Statement
- Solution Overview
- Company Description and Qualifications
- Drawings
- Walkthrough Video
- Project Schedule
- Construction Schedule
- Conclusion

Company Profile

MAE
Enterprise

Figure 1: MAE Enterprise logo

At MAE Enterprise, not only do we construct buildings with top-performing security, accessibility, and necessities, we make a second home for students and faculty.

Company Profile

Alex Duan

Head of Project Management

- BS IDM
- English and Chinese
- Skills: Microsoft Project
- Digital Media

Malek Gallichio

Head of Finance

- BS Civil Engineering
- English, French, and Arabic
- Built a school building

Elizabeth Vazquez

Head of Design

- BS Computer Science
- English and Spanish
- Skills: Autodesk Revit
- Art commissions

Figure 2: Alex's Digital Art

Figure 3: Malek's School Bathroom

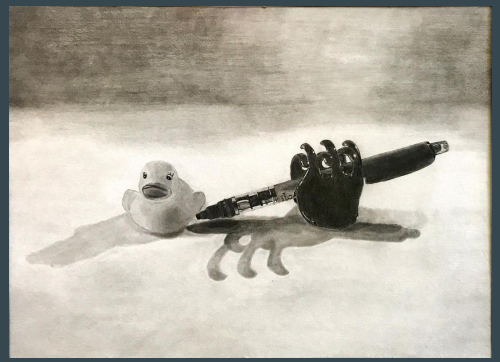

Figure 4: Elizabeth's Commissioned Artwork

Problem Statement

- MAE Enterprise has been requested to redesign the old Jacobs Building for NYU in order to modernize the building and provide sophisticated technology to NYU students

- MAE Enterprises will use engineering ingenuity to provide NYU students with the best technology and most advanced construction materials in order to ensure a quality building

  "You MAE rest assured when MAE is on the job!"

Project Description

- Uses modern technology
  - Has new classrooms:
    - Ten 40 student classrooms
    - Eight 20 student classrooms
    - Two 100 student classrooms
  - 1 Recreational space
  - Dorm space for 100 students
  - 1 Facility of choice
    - Theater

Figure 5: College classrooms

Figure 6: Theater

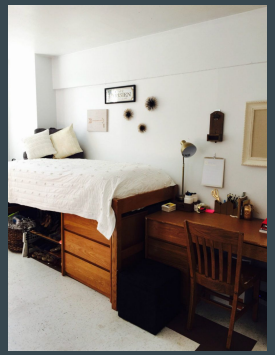

Figure 7: Dorm

Solution Overview

Design with Autodesk Revit 2019, a computer aided design (CAD) tool.

Organize floors logically

- Lower floors: welcoming spaces, exhibitions, administrative offices
- Middle floors: classrooms
- Upper floors: dorms and facilities to support students

Product Walkthrough: Initial Sketches

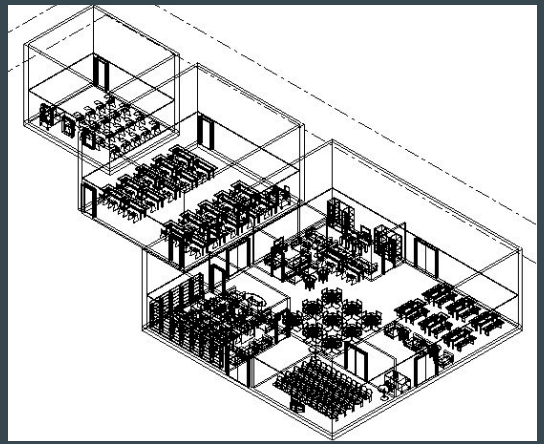

Figure 8: Milestone 1 Sketch

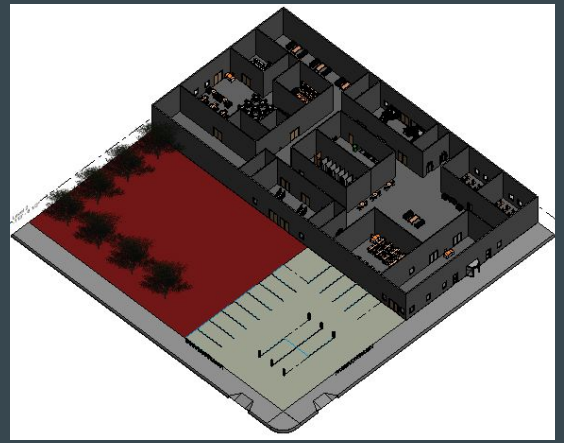

Figure 9: Milestone 2 Sketch

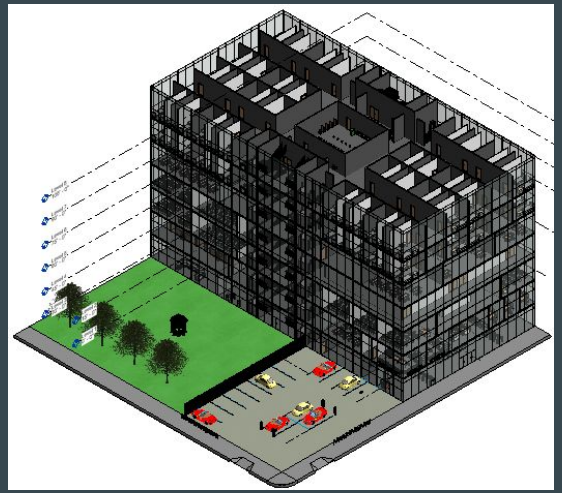

Figure 10: Milestone 3 Sketch

Product Walkthrough: Final Building Views

Figure 11: Final isometric view

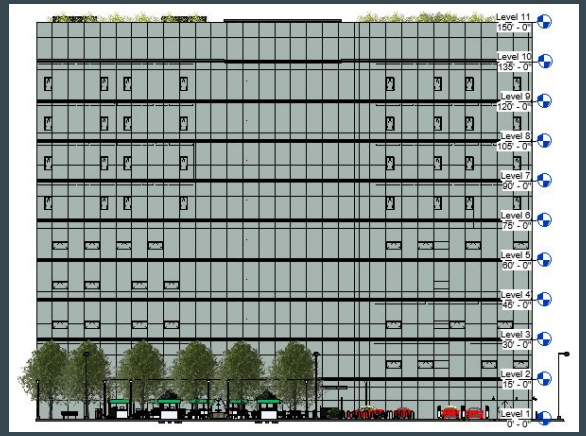

Figure 12: Most detailed side elevation

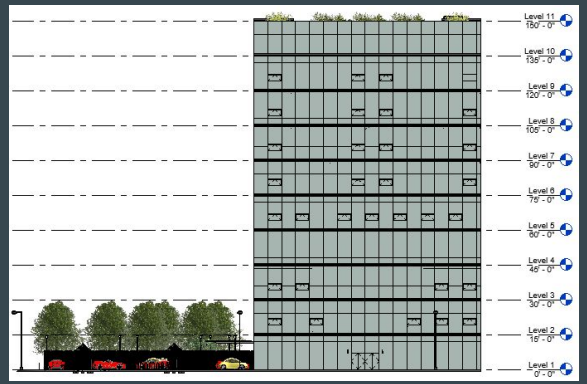

Figure 13: Front elevation

Product Walkthrough: Lower Floors

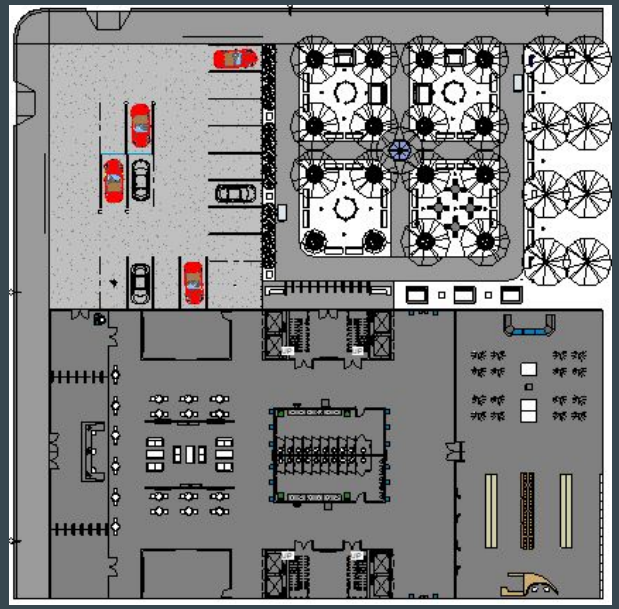

Figure 14: First floor and site plan

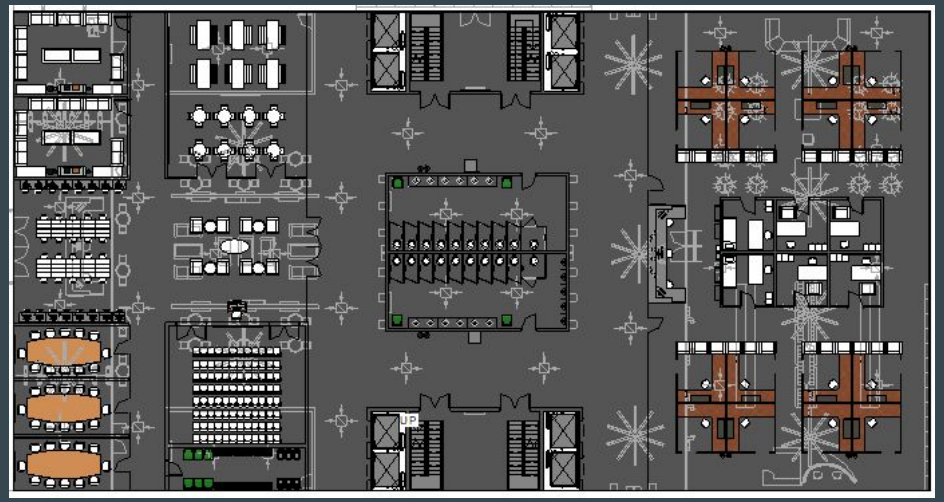

Figure 15: Second floor plan

Product Walkthrough: Middle Floors

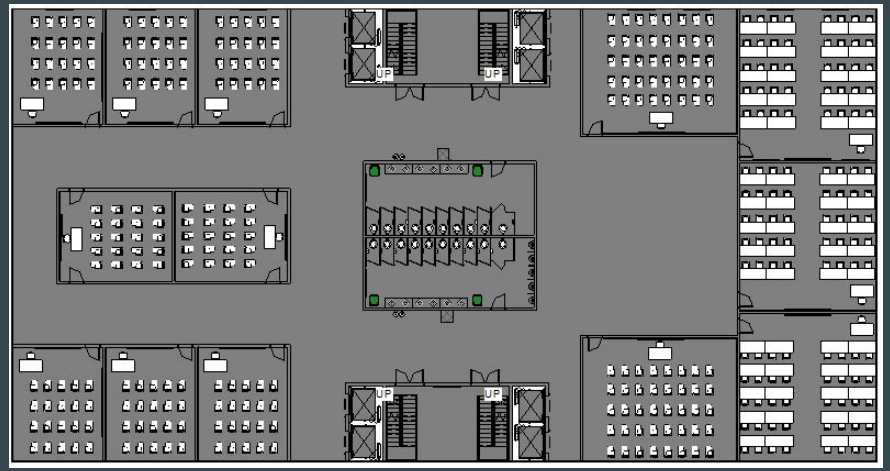

Figure 16: Third floor plan

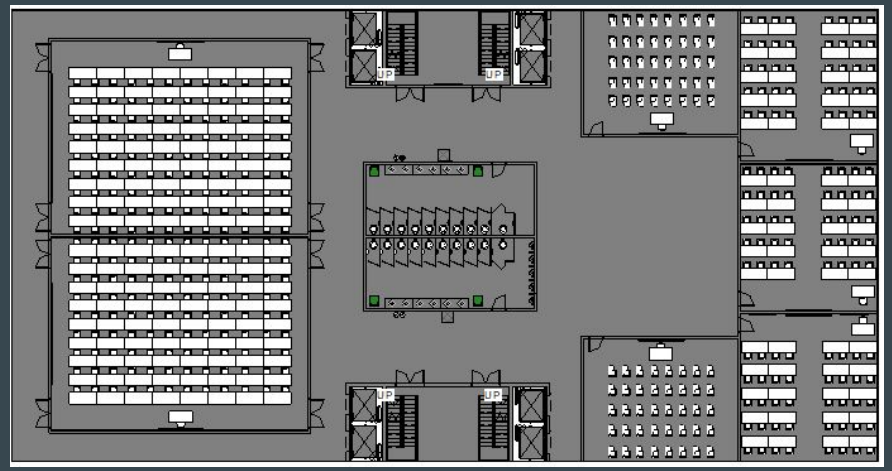

Figure 17: Fourth floor plan

Product Walkthrough: Upper Floors

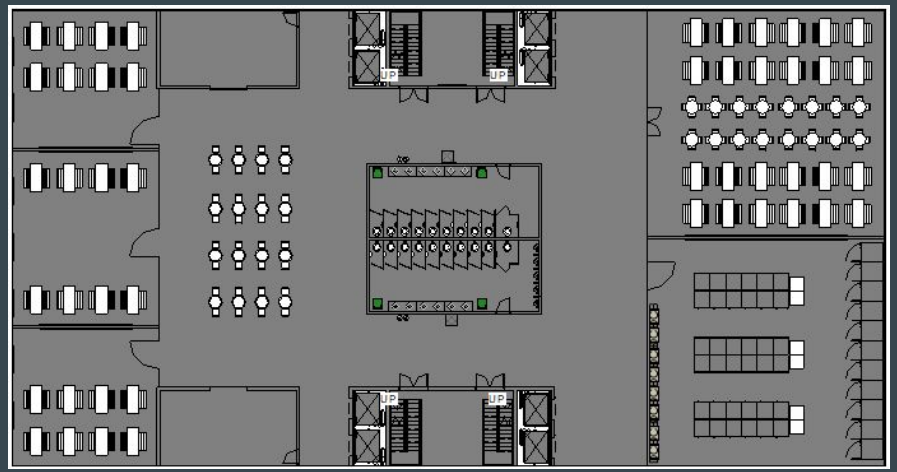

Figure 18: Fifth floor plan

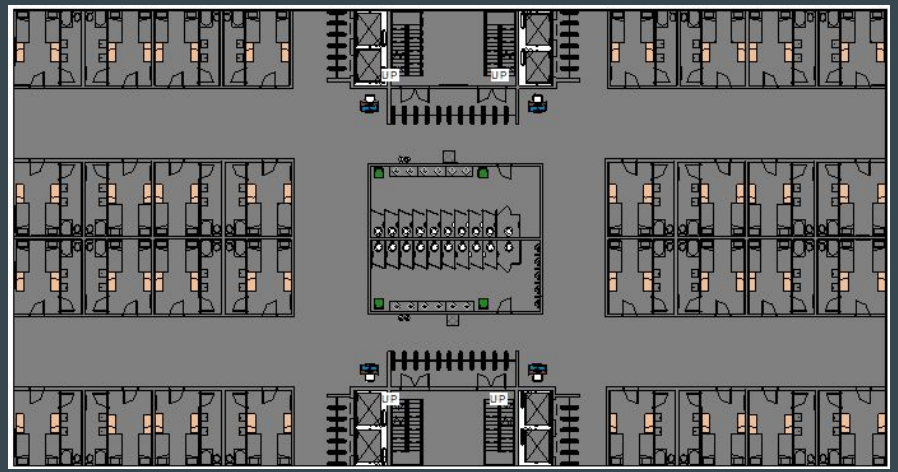

Figure 19: Sixth, seventh, eighth, and ninth floors' plan

Product Walkthrough: Roof

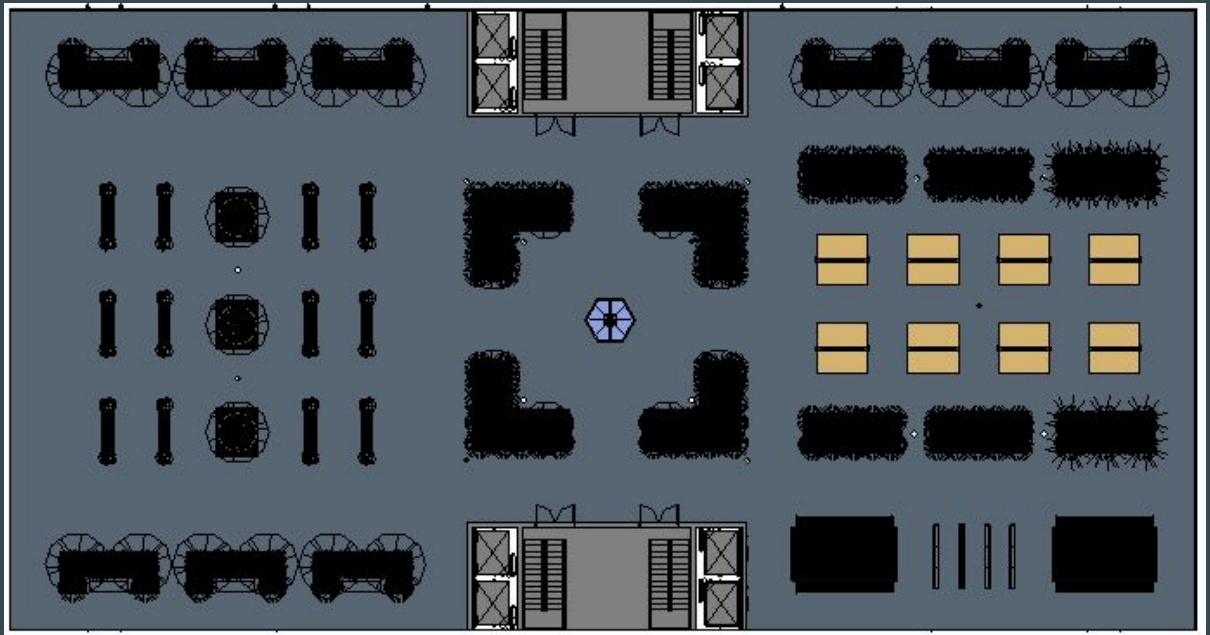

Figure 20: Roof plan

Product Walkthrough: Plumbing and Electrical Systems

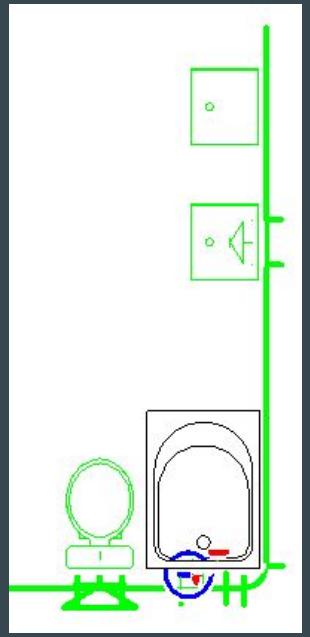

Figure 21: Plumbing plan for dorm

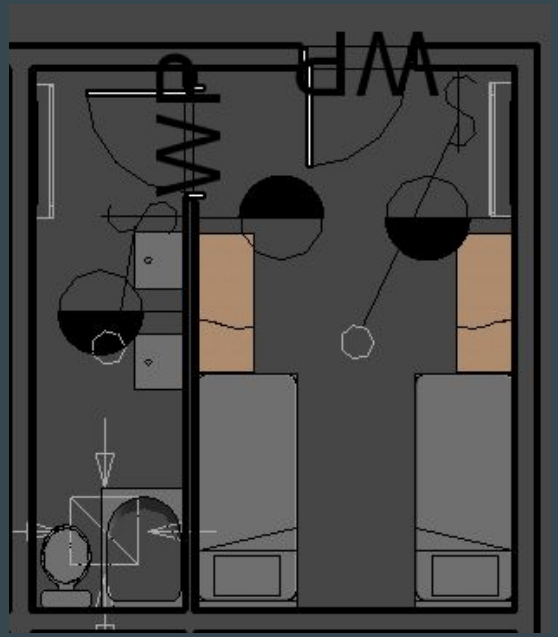

Figure 22: Electrical plan for dorm

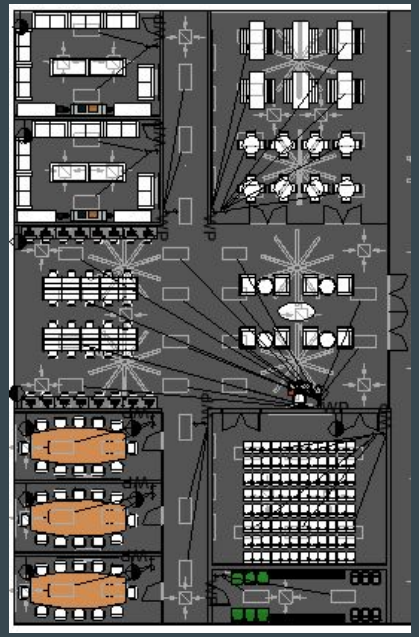

Figure 23: Electrical plan for recreational space

Product Walkthrough: Electrical System

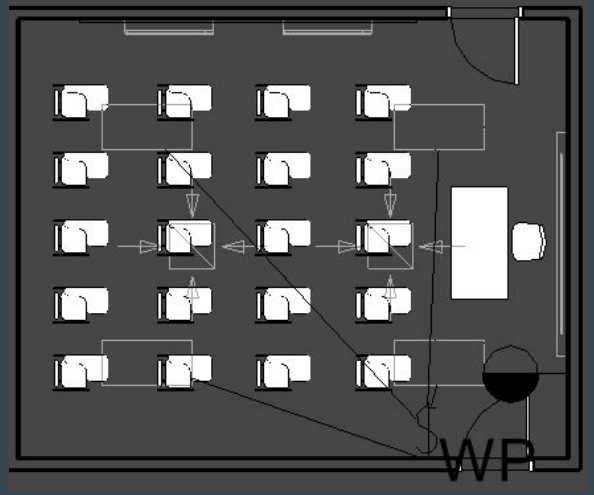

Figure 24: Electrical plan for 20-student classroom

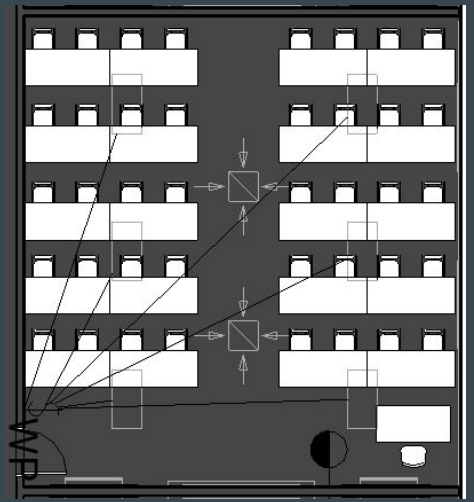

Figure 25: Electrical plan for 40-student classroom

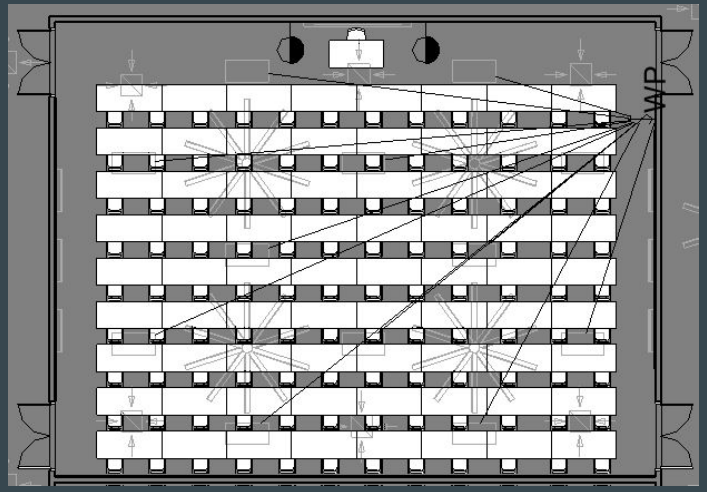

Figure 26: Electrical plan for 100-student classroom

LEED Accreditation

Gold LEED certification

- Location and Transportation
  - Parking lot
  - Spaces for vendors
- Indoor Environmental Quality
  - Return diffusers
  - Window blinds
  - Ceiling fans
  - Acoustic wall panels

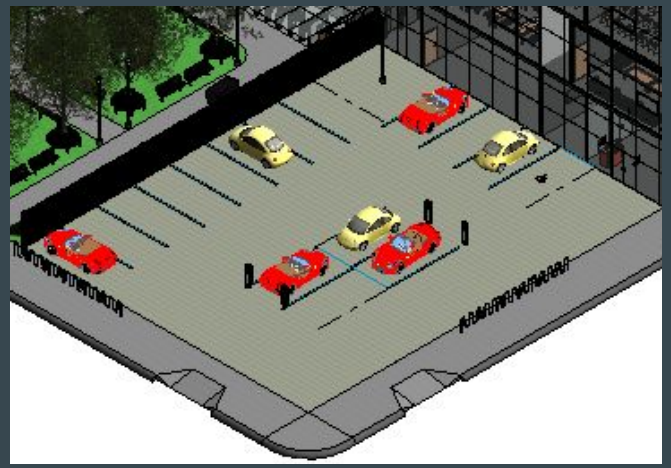

Figure 27: Parking Lot

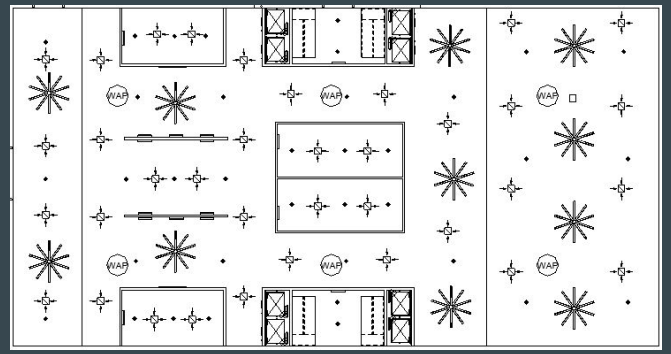

Figure 28: First floor ceiling plan

LEED Accreditation

- Sustainable Sites
  - Park
  - Rainwater retention pond
  - Rooftop garden
  - Street lights
  - Security cameras
- Innovation
  - Wireless access points
  - Exit door scanners
  - Fire pull boxes/emergency fire contact
  - Vertical garden/rotating garden carousel

Figure 29: Park space

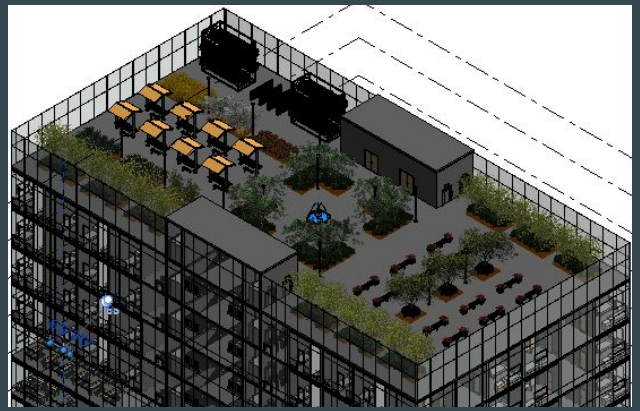

Figure 30: Rooftop garden

Walkthrough Video

Figure 31: Building Walkthrough

Cost Estimate

Table 1: Cost Estimate

Category	Material	Unit	Units Needed	Cost/Unit	Total Cost
	Steel	sq/ft	240,000	$9.75	$2,340,000
	Concrete	sq/ft	240,000	$5	$1,200,000
Foundation	Bricks	unit	2700000	$0.55	$1,485,000
	Electrical System	sq/ft	240000	$8	$1,920,000
	Security System	n/a	1	$20,000	$20,000
Electrical	Elevators	unit	8	$35,000	$280,000
	Thermal Protection	sq/ft	40800	$1.19	$48,552
	Windows	units	190	$150	$28,500
	Paint	gallons	156	$30	$4,680
	Furniture	n/a	1	$100,000	$100,000
Finishing	Educational Supplies	n/a	1	$500,000	$500,000
Miscellaneous	Miscellaneous	n/a	1	$1,500,000	$1,500,000
	Manual Labor (50)	hours	200000	$50	$10,000,000
	Head of Project Management	hours	2000	$85	$170,000
	Head of Finance	hours	1500	$80	$120,000
	Head of Design	hours	2500	$60	$150,000
Labor	Engineers (3)	hours	9000	$65	$585,000
				Total Cost	$20,451,732
				Total Cost + %20	$24,542,078

Project Schedule

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors	Resource Names
1	✓		Discuss LEED Categories and plan Design	1 day	Fri 3/5/21	Fri 3/5/21		Team
2	✓		Design Student Classrooms	3 days	Fri 3/5/21	Tue 3/9/21		Team
3	✓		Design and 3D print logo	3 days	Fri 3/5/21	Tue 3/9/21		Team
4	✓		Benchmark A	1 day	Tue 3/9/21	Tue 3/9/21		
5	✓		All Student Classrooms Revit Design	3 days	Fri 3/12/21	Tue 3/16/21		Elizabeth
6	✓		One Recreational Space Revit Deisgn	6 days	Tue 3/16/21	Tue 3/23/21		Elizabeth
7	✓		Facility of choice revit design	6 days	Tue 3/9/21	Tue 3/16/21		Team
8	✓		Electrical and Plumbing for Dorm Room	6 days	Tue 3/16/21	Tue 3/23/21		Team
9	✓		Milestone 2 Presentation	4 days	Fri 3/12/21	Wed 3/17/21		Team
10	✓		Milestone 2	1 day	Thu 3/18/21	Thu 3/18/21		
11	✓		Benchmark B	1 day	Tue 3/30/21	Tue 3/30/21		
12	✓		Milestone 3 Presentation	1 day	Thu 4/8/21	Thu 4/8/21		Team
13	✓		Revit Drawings	14 days	Thu 4/8/21	Tue 4/27/21		Team
14	✓		All Floor Plans	6 days	Thu 4/8/21	Thu 4/15/21		Team
15	✓		Plumbing System	6 days	Thu 4/15/21	Thu 4/22/21		Team
16	✓		Electrical System	6 days	Thu 4/15/21	Thu 4/22/21		Team
17	✓		Front Elevation	3 days	Thu 4/22/21	Mon 4/26/2		Team
18	✓		Most Detailed Side	3 days	Thu 4/22/21	Mon 4/26/2		Team
19	✓		Model of Building (3D Revit)	3 days	Thu 4/22/21	Mon 4/26/21		Team
20	✓		Comissioning	1 day	Tue 4/27/21	Tue 4/27/21		Team
21	✓		LEED Accredition Report	8 days	Fri 4/16/21	Tue 4/27/21		Team
22	✓		Final Presentation	5 days	Fri 4/23/21	Thu 4/29/21		Team

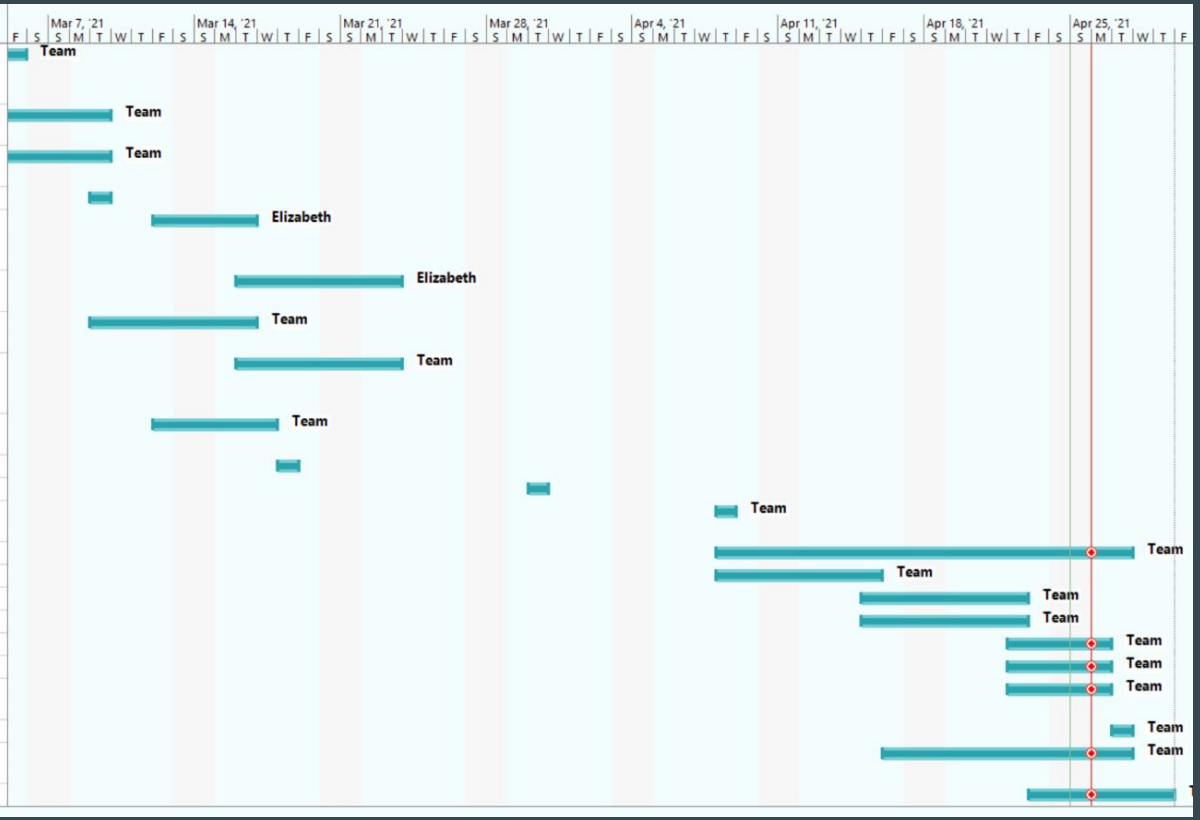

Figure 32: Final Microsoft Project schedule

Construction Schedule

ID	Task Mode	Task Name	Duration	Start	Finish	Predecessors
1		Clearing Site	7 days	Sat 5/1/21	Sat 5/8/21	
2		Raw Material Accumulation	6 days	Mon 5/3/21	Sat 5/8/21	
3		Laying Foundation	12 days	Sun 5/9/21	Mon 5/24/2	
4		Studs and Structural	4 days	Mon 5/24/21	Thu 5/27/21	
5		Floors Creation	12 days	Thu 5/27/21	Fri 6/11/21	
6		Wall Construction and Window Installation	11 days	Sun 5/30/21	Fri 6/11/21	
7		Electric Wiring and Lighting	11 days	Fri 6/11/21	Fri 6/25/21	
8		Sewage Pipelines	3 days	Fri 6/25/21	Tue 6/29/21	
9		Painting	24 days	Sun 5/30/21	Wed 6/30/2	
10		Accessories Instillation	4 days	Wed 6/30/21	Mon 7/5/21	
11		Fire Extingushers	13 days	Sun 5/30/21	Tue 6/15/21	
12		Parking Spaces	12 days	Thu 7/1/21	Fri 7/16/21	
13		Appliances	12 days	Fri 7/16/21	Sat 7/31/21	
14		Plumbing	7 days	Fri 7/16/21	Sat 7/24/21	
15		Wrap up	6 days	Sat 7/31/21	Fri 8/6/21	
16		House Cleaning	3 days	Sun 8/8/21	Tue 8/10/21	
17		Complete	3 days	Wed 8/11/2	Fri 8/13/21	

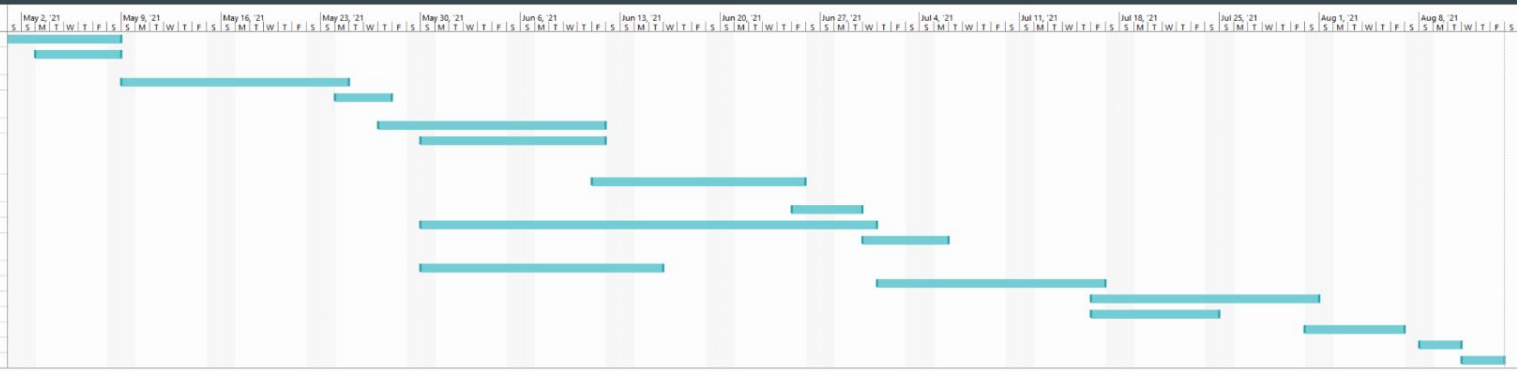

Figure 33: Construction schedule

Market and Product Viability

- Other companies in the market provide
  - Buildings
- Our company provides
  - The highest level of quality construction
  - Fair and competitive prices
  - Time efficiency
- Our designed building is
  - A classroom
  - An entertainment system
  - A home

Conclusion

- Designed a building that is modern and eco-friendly
  - Modern technology
  - Sophisticated construction material
- Gold LEED Certification
  - Building designed around street corner
  - Permits vendors to sell their goods/services
  - Green environment on the ground floor and rooftop
- At MAE Enterprise, not only do we construct buildings with top-performing security, accessibility, and necessities, we make a second home for students and faculty

"You MAE rest assured when MAE is on the job!"

Jacobs Building Redesign

MAE Enterprise

Head of Project Management Alex Duan
Head of Finance Malek Gallichio
Head of Design Elizabeth Vazquez

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